

so that stocks permanently go into a bull market. Rebalancing also underperforms a buy-and-hold strategy because rebalancing would have sold stocks that only keep going up.

Permanent regime changes sometimes occur, but they are rare.³¹ I fully agree with Reinhart and Rogoff (2011) that people too often think, “This time is different.” Two examples of true regime changes where “these times really were different” are the changing shape of the yield curve pre- and post-1933 and the pricing of out-of-the-money put options pre- and post-1987. Pre-1933 the yield curve was downward sloping, compared to its now (post-1933) normal upward-sloping shape. Academics think going off the gold standard plays a role in explaining the change (see Wood (1983)). Implied option volatilities were symmetric across strikes in the pre-1987 sample (called the *implied volatility smile*). After the 1987 crash, there has been significant negative skewness in implied volatilities (see Rubinstein (1994)), with option volatilities for low strikes much larger than option volatilities for high strikes. The implied volatility smile turned into an *implied volatility smirk*. The financial crisis in 2007 and 2008 was not a regime change. The European sovereign debt crisis since 2009 is also not a regime change. True permanent regime changes are rare.

The fact is that rebalancing is short volatility, and thus short regime changes, means that you must practice rebalancing on broad asset classes (or across factors, see chapter 14) that are extremely unlikely to undergo permanent regime change. Russian equities in 1900 were a large market and then totally disappeared, with investors receiving zero, less than two decades later. But global equities are still around more than a century later and are likely to be here for a long time, perhaps until capitalism disappears. Russian bonds also disappeared during the Russian Revolution, but global bonds did not. Global equities and global bonds have been and will continue to be with us for a long time. Rebalance with the tried and true.³²

4. Liability Hedging

4.1. LIABILITY HEDGING PORTFOLIO

Few investors are without liabilities. Even investors lacking explicit liabilities (like Norway), at least over the short term, often have implicit liabilities through stewardship expectations of stakeholders. Liabilities can be fixed, like loan payments;

³¹ Recurring regime changes, in contrast, are common. Examples include recessions versus expansions, bear markets versus bull markets, high volatility versus low volatility periods, and, more generally, bad times versus good times. Recurring regime changes are well described by regime-switching processes introduced into economics and finance by Hamilton (1989). See Ang and Timmermann (2012) for a summary.

³² If asset returns follow *Markov processes*, then you want to rebalance over assets or strategies that are *recurrent*.